

1 Summary

Model (Run-name): model1 (test)

Iteration: 21

TAC: 154866.4312300252

Took: 2.9600043296813965s

Total Time: 31.990850925445557s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

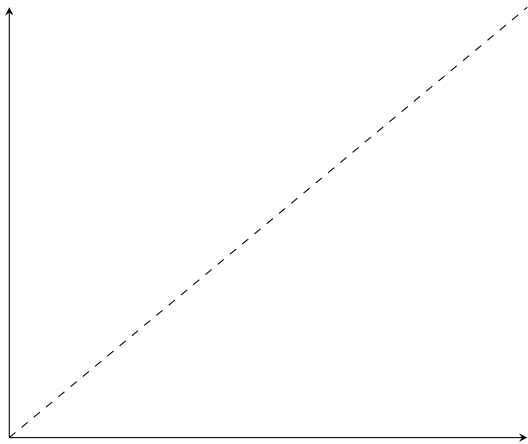
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

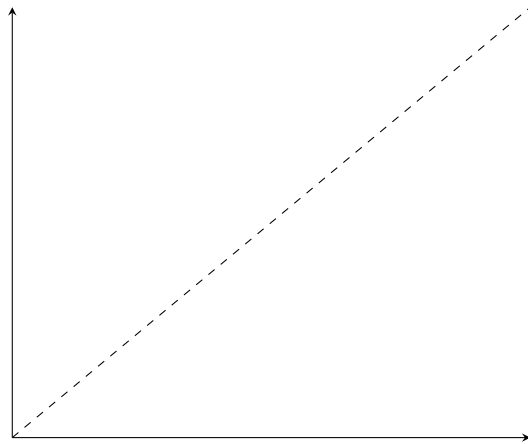
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$RecLMTD_{1,1,1}$	0.05216174207894452	$b_{1,1,1}^{H,out}$	901.2095811121504
$z_{1,2,2}$	1.0	$RecLMTD_{1,2,2}$	0.017750398843076914	$b_{1,2,2}^{H,out}$	3868.0035602394537
$z_{2,1,2}$	1.0	$RecLMTD_{2,1,2}$	0.028928374676624933	$b_{2,1,2}^{H,out}$	9372.99465964082
$z_{cu,1}$	1.0	$RecLMTD_{cu,1}$	0.014622489280066561	$b_{1,1,1}^{C,in}$	5637.792572892671
$z_{cu,2}$	1.0	$RecLMTD_{cu,2}$	0.009574839464308246	$b_{1,2,2}^{C,in}$	4550.0
$z_{hu,1}$	1.0	$RecLMTD_{hu,1}$	0.022535917706857686	$b_{2,1,2}^{C,in}$	4175.113749686419
$A_{1,1,1}^\beta$	71.00935653192329	$q_{1,1,1}$	681.9964397605463	$b_{1,1,1}^{C,out}$	6319.7890126532175
$A_{1,2,2}^\beta$	69.22655548799992	$q_{1,2,2}$	1950.0	$b_{1,2,2}^{C,out}$	6500.0
$A_{2,1,2}^\beta$	140.31685261310218	$q_{2,1,2}$	2427.005340359181	$b_{2,1,2}^{C,out}$	6602.1190900455995
$A_{cu,1}^\beta$	4.888121952381175	$q_{cu,1}$	168.0035602394537		
$A_{cu,2}^\beta$	37.76995088018704	$q_{cu,2}$	1972.994659640819		
$A_{hu,1}^\beta$	13.266498143734903	$q_{hu,1}$	490.9982198802727		
$A_{1,1,1}$	71.00935653192329	$t_{1,1}^{(H)}$	650.0		
$A_{1,2,2}$	69.22655548799992	$t_{1,2}^{(H)}$	581.8003560239454		
$A_{2,1,2}$	140.31685261310218	$t_{1,3}^{(H)}$	386.80035602394537		
$A_{cu,1}$	4.888121952381175	$t_{2,1}^{(H)}$	590.0		
$A_{cu,2}$	37.76995088018704	$t_{2,2}^{(H)}$	590.0		
$A_{hu,1}$	13.266498143734903	$t_{2,3}^{(H)}$	468.649732982041		
$dt_{1,1,1}$	32.733214658684844	$t_{1,1}^{(C)}$	617.2667853413152		
$dt_{1,1,2}$	10.0	$t_{1,2}^{(C)}$	571.8003560239454		
$dt_{1,2,2}$	81.80035602394537	$t_{1,3}^{(C)}$	410.0		
$dt_{1,2,3}$	36.80035602394537	$t_{2,1}^{(C)}$	500.0		
$dt_{2,1,2}$	18.19964397605463	$t_{2,2}^{(C)}$	500.0		
$dt_{2,1,3}$	58.649732982040945	$t_{2,3}^{(C)}$	350.0		
$dt_{cu,1}$	66.80035602394537	$t_{1,1,1}^H$	370.0		
$dt_{cu,2}$	148.64973298204094	$t_{1,2,2}^H$	386.80035602394537		
$dt_{hu,1}$	62.733214658684844	$t_{2,1,2}^H$	468.64973298204103		
$f_{1,1,1}^H$	2.4357015705733795	$t_{1,1,1}^C$	640.8857765174175		
$f_{1,2,2}^H$	10.0	$t_{1,2,2}^C$	500.0		
$f_{2,1,2}^H$	20.0	$t_{2,1,2}^C$	648.3341506855953		
$f_{1,1,1}^C$	9.86102242273973	$b_{1,1,1}^{H,in}$	1583.2060208726966		
$f_{1,2,2}^C$	13.0	$b_{1,2,2}^{H,in}$	5818.003560239454		
$f_{2,1,2}^C$	10.183204267527852	$b_{2,1,2}^{H,in}$	11800.0		

3 *RecLMTD* Tangent Points

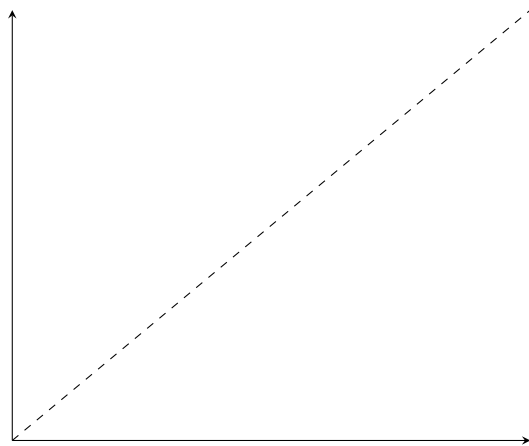
- $(1, 1, 1)$
 - Value: 0.052162
 - Error: 0.000000049642854
 - Relative error: 0.000000951709150
- $(1, 2, 2)$
 - Value: 0.017750
 - Error: 0.000000136185332
 - Relative error: 0.000007672181786
- $(2, 1, 2)$
 - Value: 0.028928
 - Error: 0.000000634355812
 - Relative error: 0.000021928017363
- $cu, 1$
 - Value: 0.014622
 - Error: 0.000000022527910
 - Relative error: 0.000001540632045
- $cu, 2$
 - Value: 0.009575
 - Error: 0.000000495156075
 - Relative error: 0.000051711621012
- $hu, 1$
 - Value: 0.022536
 - Error: 0.000000630346225
 - Relative error: 0.000027969954551



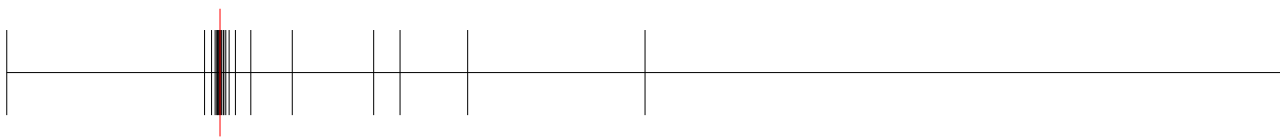
(a) *RecLMTD* - (1, 1, 1)



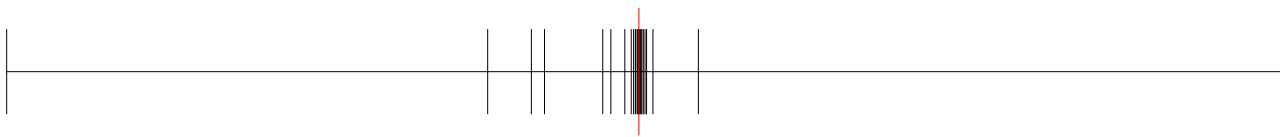
(b) *RecLMTD* - (1, 2, 2)



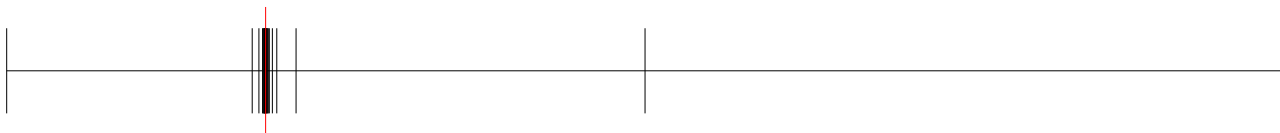
(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - cu, 1



(b) *RecLMTD* - cu, 2



(a) *RecLMTD* - hu, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 1583.2060208726966
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5818.003560239454
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 11800.0
 - Absolute error: 0.0
 - Relative error: 0.0

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 901.2095811121504
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 3868.0035602394537
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 9372.99465964082
 - Absolute error: 0.0
 - Relative error: 0.0

4.3 $b^{C,\text{in}}$

- (1, 1, 1)
 - Value: 5637.792572892671
 - Absolute error: 0.7435591900148211
 - Relative error: -0.00013187096306504917
- (1, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
 - Value: 4175.113749686419
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 6319.7890126532175
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 6602.1190900455995
 - Absolute error: 0.0
 - Relative error: 0.0



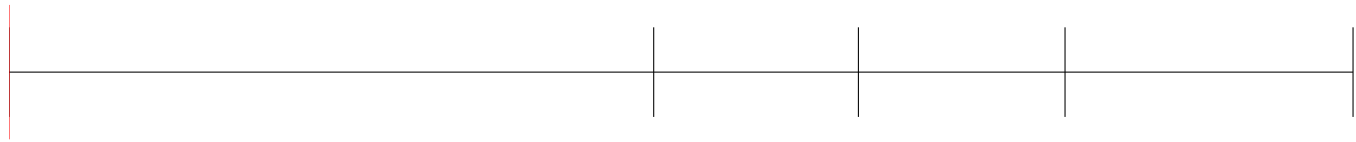
(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



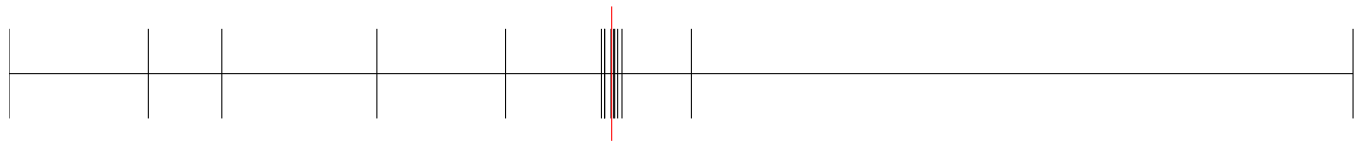
(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



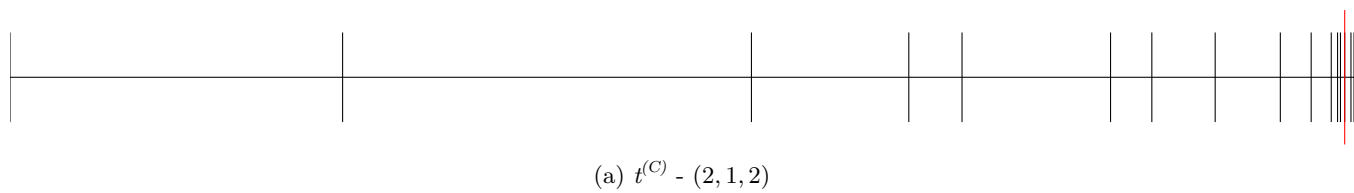
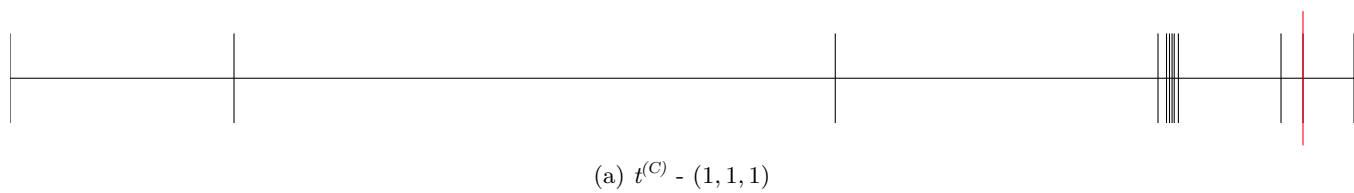
(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$



5 Area Breakpoints

5.1 A

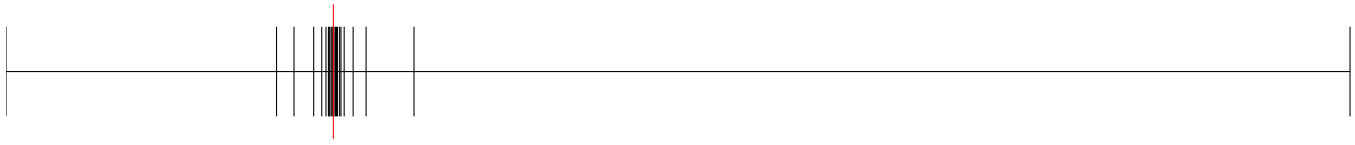
- $(1, 1, 1)$
 - Value: 71.009357
 - Error: 0.138888247172801
 - Relative error: 0.001952096606234
- $(1, 2, 2)$
 - Value: 69.226555
 - Error: 0.0000000000000043
 - Relative error: 0.000000000000001
- $(2, 1, 2)$
 - Value: 140.316853
 - Error: 0.101787043057840
 - Relative error: 0.000724882702945

5.2 A_{CU}

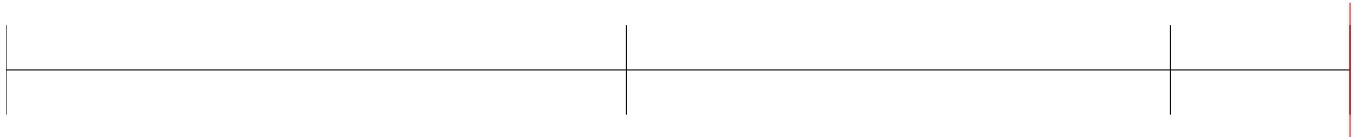
- $cu, 1$
 - Value: 4.888122
 - Error: 0.025138564847682
 - Relative error: 0.005116473014108
- $cu, 2$
 - Value: 37.769951
 - Error: 0.012263379809617
 - Relative error: 0.000324580759752

5.3 A_{HU}

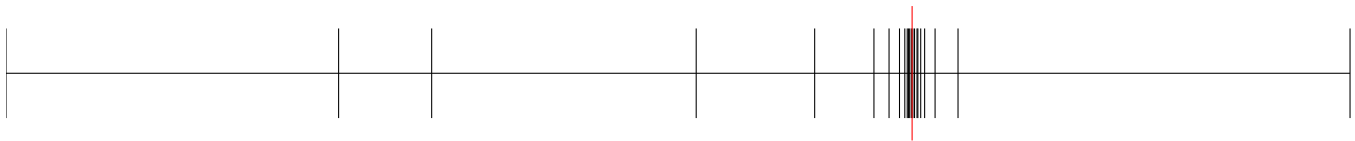
- $hu, 1$
 - Value: 13.266498
 - Error: 0.011616429187626
 - Relative error: 0.000874855321049



(a) $A^\beta - (1, 1, 1)$



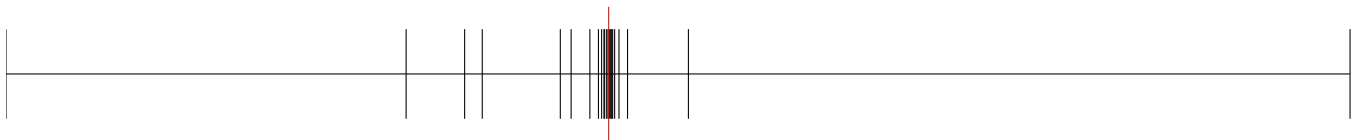
(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$

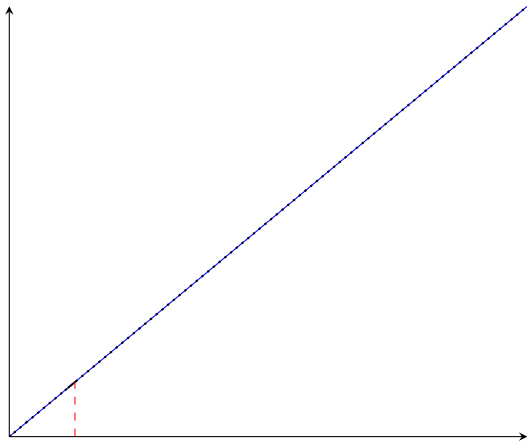


(a) $A^\beta - hu, 1$

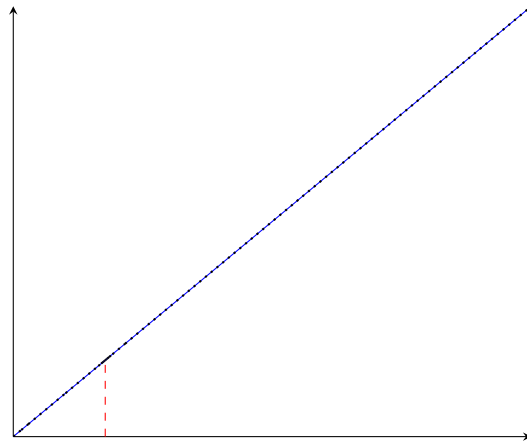
6 A^β Breakpoints

- (1, 1, 1)
 - q : 71.009357
 - $\hat{q}^\beta$: 71.009357
 - q^β : 71.009357
 - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 67.27362048222933, 69.14107665413653, 69.42777867315647, 69.89902342903967, 70.22290217272871, 70.30181758862284, 70.46247894104941, 70.50425945122109, 70.60541450524272, 70.95870595008942, 71.22933662920877, 71.2616536557244, 71.33750386361461, 71.6637831223933, 71.88123217857378, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1, 2, 2)
 - q : 69.226555
 - $\hat{q}^\beta$: 69.226555
 - q^β : 69.226555
 - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 66.17325582523148, 67.27673099173637, 67.91344539650363, 68.10969007738701, 68.66676327863097, 69.0314457108091, 69.34502840170047, 69.41355730879985, 70.08419437796795, 70.37078617087542, 70.94081287188642, 72.25348351595485, 74.6496383881616, 84.1468297226002, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2, 1, 2)
 - q : 140.316853
 - $\hat{q}^\beta$: 140.316853
 - q^β : 140.316853
 - Breakpoints: [0, 9.898970403313276, 28.333333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 118.88765652962957, 128.52021081477858, 131.69760457629607, 132.12228195956155, 135.81963163113545, 137.02923097052596, 139.47623233772285, 140.7008934199032, 142.4338700260237, 144.67005099128556, 144.70843093515643, 147.07009017600456, 147.48374761404892, 149.84689443310864, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 4.888122
 - $\hat{q}^\beta$: 4.888122
 - q^β : 4.888122
 - Breakpoints: [0, 2.5450346292088537, 2.9819181007907165, 3.1809419553840232, 3.9948823402911255, 4.269280092716329, 4.432280503220532, 4.613047202566896, 4.707108999269609, 4.825601156120121, 4.849719301830286, 4.974201802143407, 4.983938626310074, 4.993068227518925, 5.0418124100090695, 5.081618324442152, 7.702824927089129, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]
 - Error: 0.0000000000000000

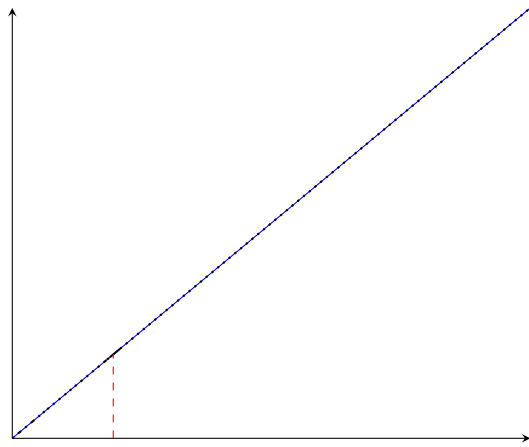
- Relative error: 0.0000000000000000
- $cu, 2$
 - q : 37.769951
 - $\hat{q}^\beta$: 37.769951
 - q^β : 37.769951
 - Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717, 36.670233878040335, 37.21616748032837, 37.443498367229196, 37.47875108132638, 37.554741031910254, 37.612770977560544, 37.6551299822246, **37.682669717644174**, **37.78573239590169**, 37.88148462263016, 37.9018730666985, 37.9454585315849, 38.10872593446926, 38.21652191748154, 38.88483572219945, 285.4001551947793]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $hu, 1$
 - q : 13.266498
 - $\hat{q}^\beta$: 13.266498
 - q^β : 13.266498
 - Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, 12.917705220652834, 13.104034300095861, 13.161020122653593, 13.189852638760229, 13.192252946941991, 13.213749911872785, 13.218222920629291, 13.22717920580809, **13.244985069124171**, **13.269097009439808**, 13.30871858348944, 13.32330648750218, 13.32663103908035, 13.346967780023082, 13.439500729482463, 13.468285806082731, 14.09178614751222, 237.3002543523117]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



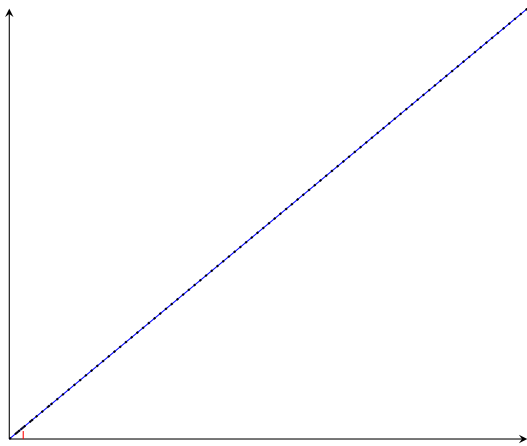
(a) $q^\beta - (1, 1, 1)$



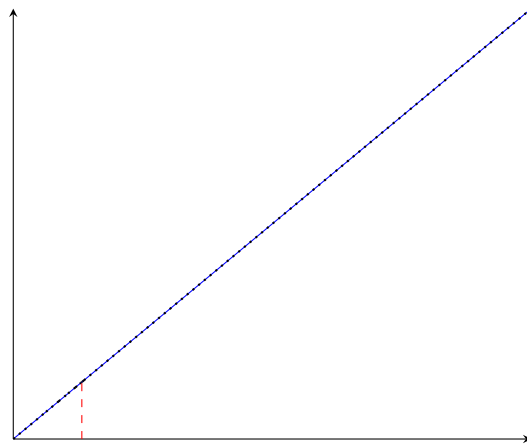
(b) $q^\beta - (1, 2, 2)$



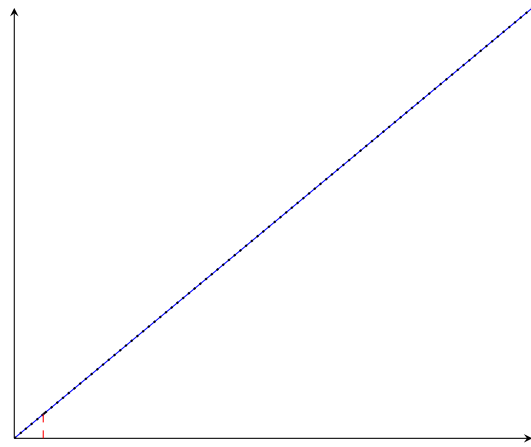
(c) $q^\beta - (2, 1, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	-0.0	$f_{2,2,1}^C$	0.0	$b_{2,2,2}^{H,out}$	0.0
$z_{1,2,1}$	-0.0	$f_{2,2,2}^C$	0.0	$b_{1,1,2}^{C,in}$	1974.8862503135806
$z_{2,1,1}$	0.0	$RecLMTD_{1,1,2}$	0.0	$b_{1,2,1}^{C,in}$	6500.0
$z_{2,2,1}$	-0.0	$RecLMTD_{1,2,1}$	0.008079076877753476	$b_{2,1,1}^{C,in}$	2939.2127674665107
$z_{2,2,2}$	-0.0	$RecLMTD_{2,1,1}$	0.029166666666666688	$b_{2,2,1}^{C,in}$	0.0
$z_{hu,2}$	-0.0	$RecLMTD_{2,2,1}$	0.004513888888888885	$b_{2,2,2}^{C,in}$	0.0
$A_{1,1,2}^\beta$	0.0	$RecLMTD_{2,2,2}$	0.0	$b_{1,1,2}^{C,out}$	1974.8862503135815
$A_{1,2,1}^\beta$	0.0	$RecLMTD_{hu,2}$	0.005549329826813255	$b_{1,2,1}^{C,out}$	6500.0
$A_{2,1,1}^\beta$	0.0	$q_{1,1,2}$	0.0	$b_{2,1,1}^{C,out}$	2939.2127674665107
$A_{2,2,1}^\beta$	0.0	$q_{1,2,1}$	0.0	$b_{2,2,1}^{C,out}$	0.0
$A_{2,2,2}^\beta$	0.0	$q_{2,1,1}$	0.0	$b_{2,2,2}^{C,out}$	0.0
$A_{hu,2}^\beta$	0.0	$q_{2,2,1}$	0.0		
$A_{1,1,2}$	0.0	$q_{2,2,2}$	0.0		
$A_{1,2,1}$	0.0	$q_{hu,2}$	0.0		
$A_{2,1,1}$	0.0	$t_{1,1,2}^H$	370.0		
$A_{2,2,1}$	0.0	$t_{1,2,1}^H$	650.0		
$A_{2,2,2}$	0.0	$t_{2,1,1}^H$	590.0		
$A_{hu,2}$	0.0	$t_{2,2,1}^H$	590.0		
$dt_{1,1,3}$	240.0	$t_{2,2,2}^H$	459.46103212427704		
$dt_{1,2,1}$	150.0	$t_{1,1,2}^C$	410.0		
$dt_{2,1,1}$	24.166666666666668	$t_{1,2,1}^C$	500.0		
$dt_{2,2,1}$	220.0	$t_{2,1,1}^C$	549.2245202127368		
$dt_{2,2,2}$	220.0	$t_{2,2,1}^C$	350.0		
$dt_{2,2,3}$	240.0	$t_{2,2,2}^C$	440.0112035847521		
$dt_{hu,2}$	180.0	$b_{1,1,2}^{H,in}$	0.0		
$f_{1,1,2}^H$	0.0	$b_{1,2,1}^{H,in}$	4916.793979127304		
$f_{1,2,1}^H$	7.5642984294266205	$b_{2,1,1}^{H,in}$	11800.0		
$f_{2,1,1}^H$	20.0	$b_{2,2,1}^{H,in}$	0.0		
$f_{2,2,1}^H$	0.0	$b_{2,2,2}^{H,in}$	0.0		
$f_{2,2,2}^H$	0.0	$b_{1,1,2}^{H,out}$	0.0		
$f_{1,1,2}^C$	4.816795732472148	$b_{1,2,1}^{H,out}$	4916.793979127303		
$f_{1,2,1}^C$	13.0	$b_{2,1,1}^{H,out}$	11800.0		
$f_{2,1,1}^C$	5.13897757726027	$b_{2,2,1}^{H,out}$	0.0		